

E-COMMERCE SALES DATA VISUALIZATION AND INSIGHTS USING PYTHON

Project Report

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1. Introduction

The current project will explore how E-Commerce sales data can be visualized and analyzed using Python, Pandas, Matplotlib, and Seaborn in Google Colab.

Data visualization is an important part of data analysis since it enables the conversion of numbers into meaningful images. With the help of different charts and graphs, the analyst will be able to detect the presence of patterns, trends, and relationships in the data set under investigation.

In our case study, the data set was initially explored and cleaned prior to the creation of visualizations. Various charts have been developed to explore sales by product, payment type, order status, referral traffic sources, monthly sales dynamics, and the company's performance in terms of revenue.

2. Objectives

- To understand the structure of the E-Commerce dataset
- To identify and handle missing values
- To perform data preparation before visualization
- To create meaningful charts and graphs using Python
- To analyze product sales performance
- To study customer payment preferences
- To identify sales trends and revenue patterns
- To generate business insights through data visualization

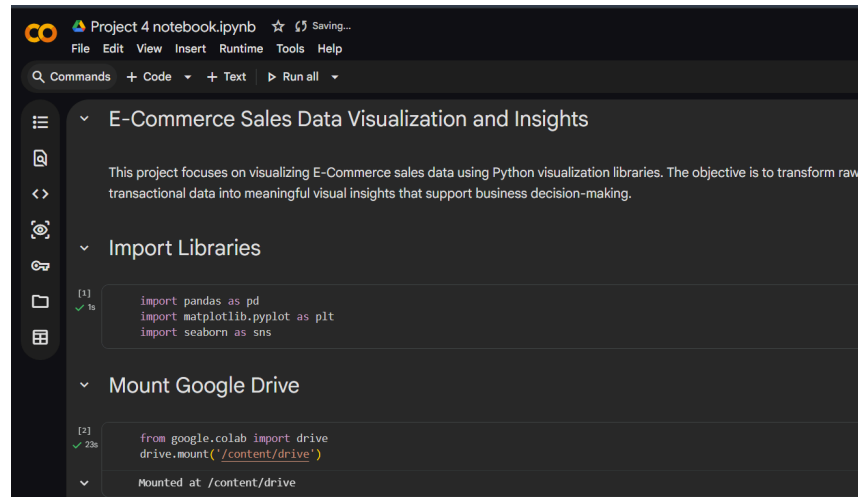
3. Dataset Description

The dataset used in this project contains E-Commerce sales transaction records. It includes information related to products, customers, payment methods, order status, referral sources, transaction values, and sales activities.

The dataset was analyzed to understand sales performance and customer behavior through visual representations. Before visualization, the data was inspected and cleaned to ensure accuracy and consistency.

4. Data Preparation

Importing Libraries



The screenshot shows a Google Colab notebook titled "Project 4 notebook.ipynb". The notebook has two code cells. The first cell, labeled "[1] 1s", contains the following code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

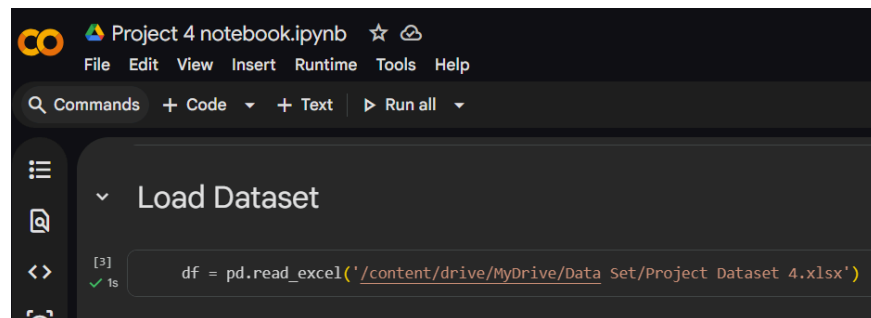
. The second cell, labeled "[2] 23s", contains the following code:

```
from google.colab import drive
drive.mount('/content/drive')
```

. Below the code, it says "Mounted at /content/drive".

The required Python libraries were imported to perform data analysis and visualization tasks. Pandas was used for data manipulation, while Matplotlib and Seaborn were used to generate charts and graphs.

Loading the Dataset

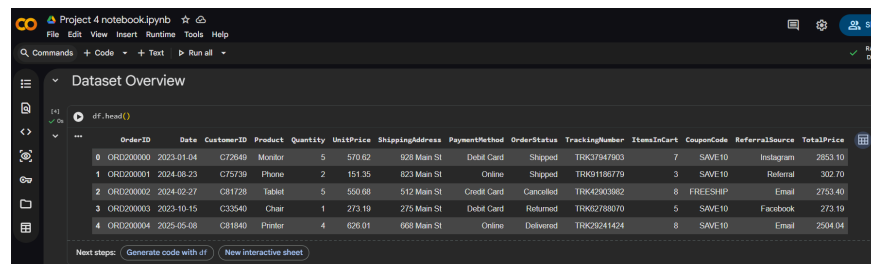


The screenshot shows a Google Colab notebook titled "Project 4 notebook.ipynb". The notebook has one code cell, labeled "[3] 1s", with the following code:

```
df = pd.read_excel('/content/drive/MyDrive/Data Set/Project Dataset 4.xlsx')
```

The E-Commerce dataset was loaded into Google Colab using Pandas. This allowed the dataset to be inspected and prepared for further analysis and visualization.

Dataset Inspection



The screenshot shows a Google Colab notebook titled "Project 4 notebook.ipynb". The notebook has one code cell, labeled "[4] 0s", with the following code:

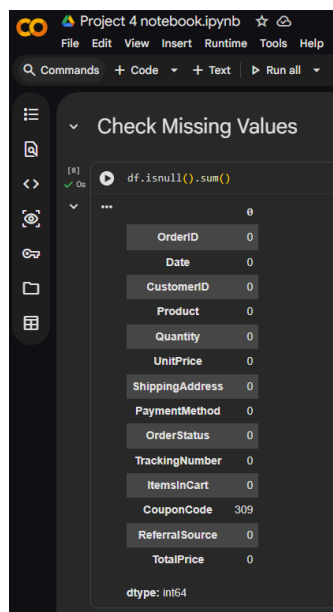
```
df.head()
```

. Below the code, a "Dataset Overview" table is displayed, showing the first five rows of the dataset.

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.10
1	ORD200001	2024-06-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91196779	3	SAVE10	Referral	302.70
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903882	8	FREESHIP	Email	2753.40
3	ORD200003	2022-10-15	C33540	Chair	1	272.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	272.19
4	ORD200004	2023-05-08	C81840	Printer	4	626.01	688 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04

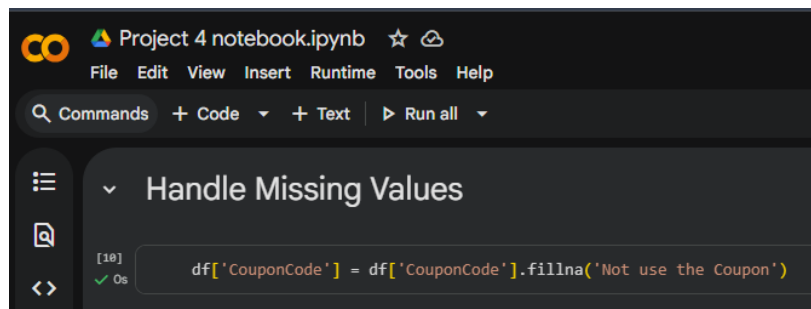
The first few records of the dataset were displayed to understand its structure, column names, and available information.

Missing Value Detection



The dataset was checked for missing values to identify incomplete records that could affect the accuracy of the visualizations and analysis.

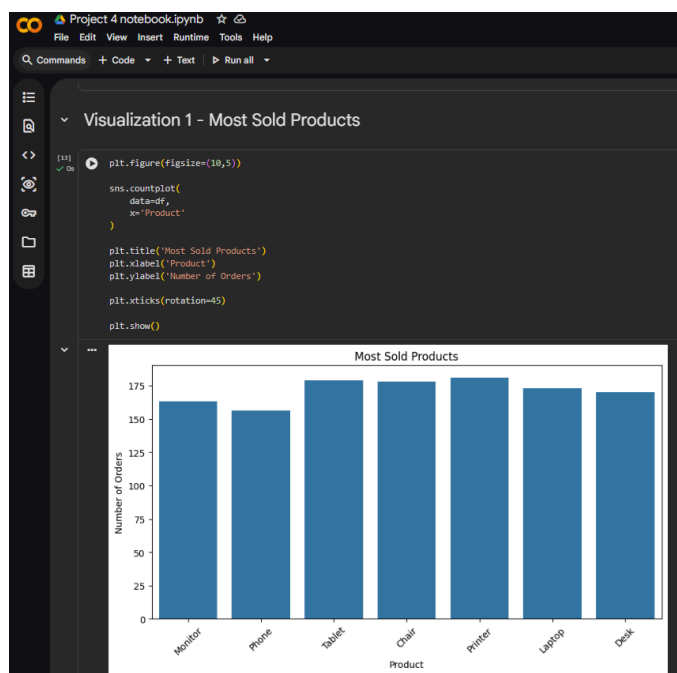
Handling Missing Values



Missing values were handled appropriately to improve data quality and ensure reliable visualization results.

5 Data Visualization

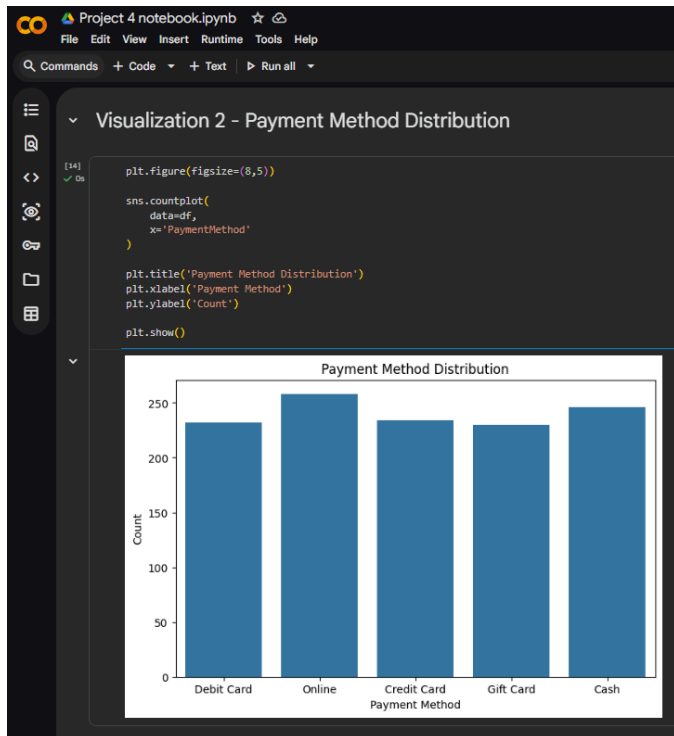
Most Sold Products Analysis



A visualization was created to identify the products with the highest number of sales. This analysis helped determine which products were most popular among customers.

The chart revealed that some products received significantly higher sales compared to others, indicating stronger customer demand.

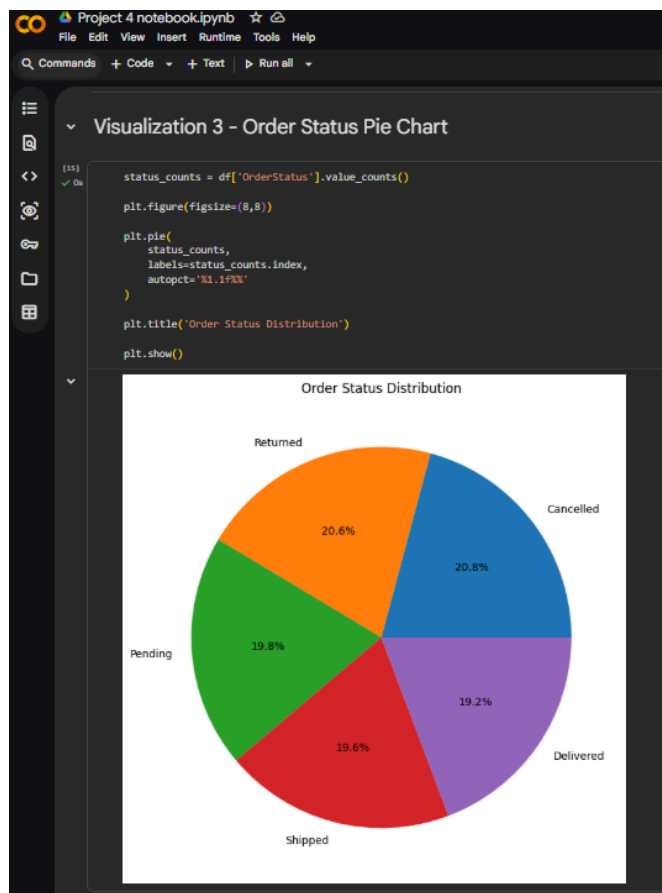
Payment Method Distribution



A chart was created to analyze the payment methods used by customers during transactions.

The analysis showed customer preferences for specific payment methods, highlighting the most commonly used transaction options.

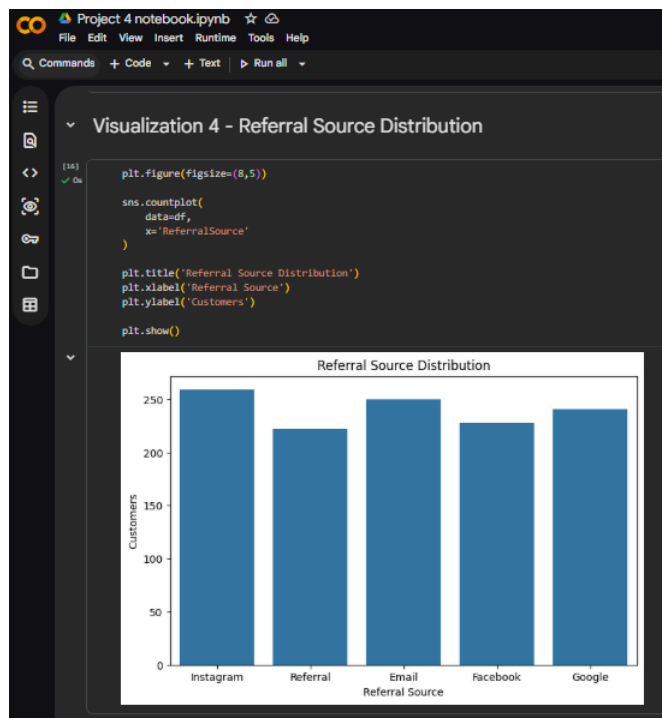
Order Status Distribution



The order status data was visualized to understand the distribution of completed, pending, and cancelled orders.

Most orders were successfully completed, while a smaller portion consisted of pending or cancelled transactions.

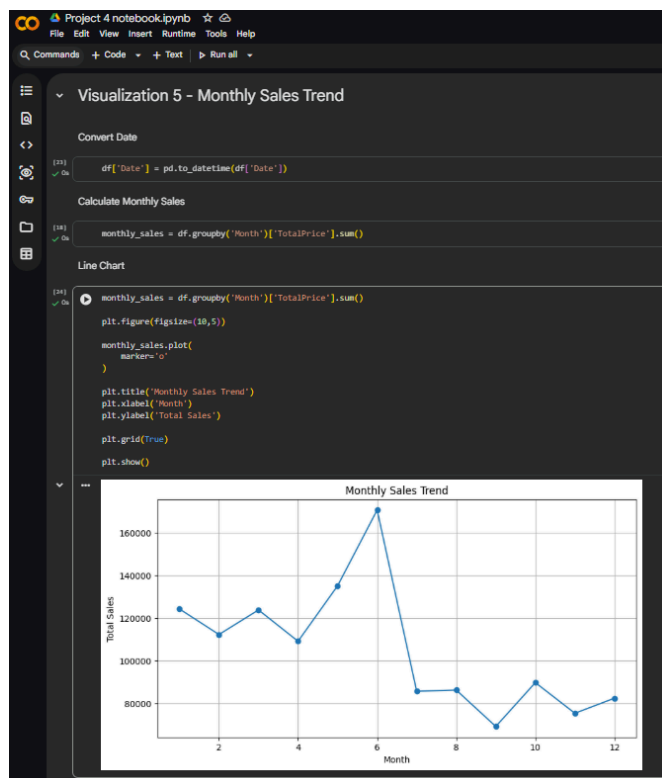
Referral Source Analysis



Referral source information was visualized to determine how customers reached the platform.

Some referral channels generated more customer activity than others, demonstrating their effectiveness in attracting users.

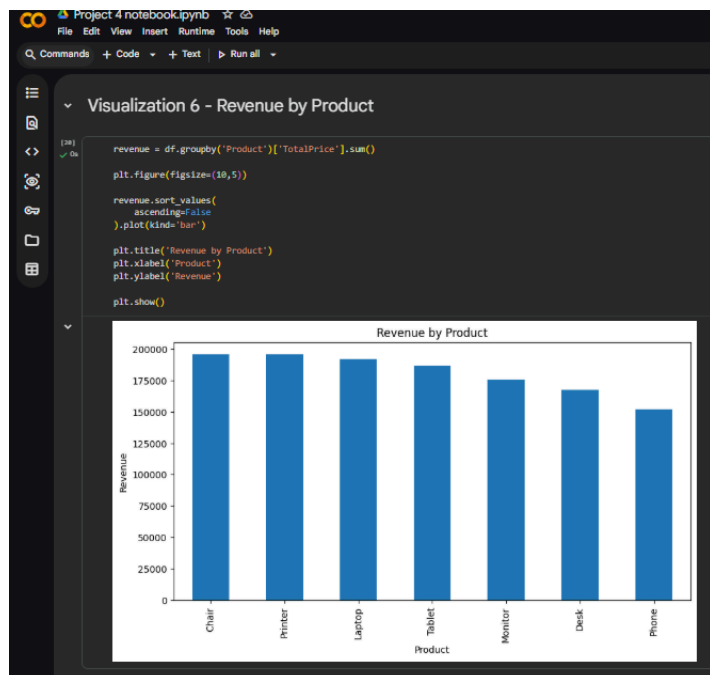
Monthly Sales Trend Analysis



A line chart was created to analyze monthly sales performance and identify trends over time.

The visualization showed fluctuations in sales across different periods and helped identify high-performing months.

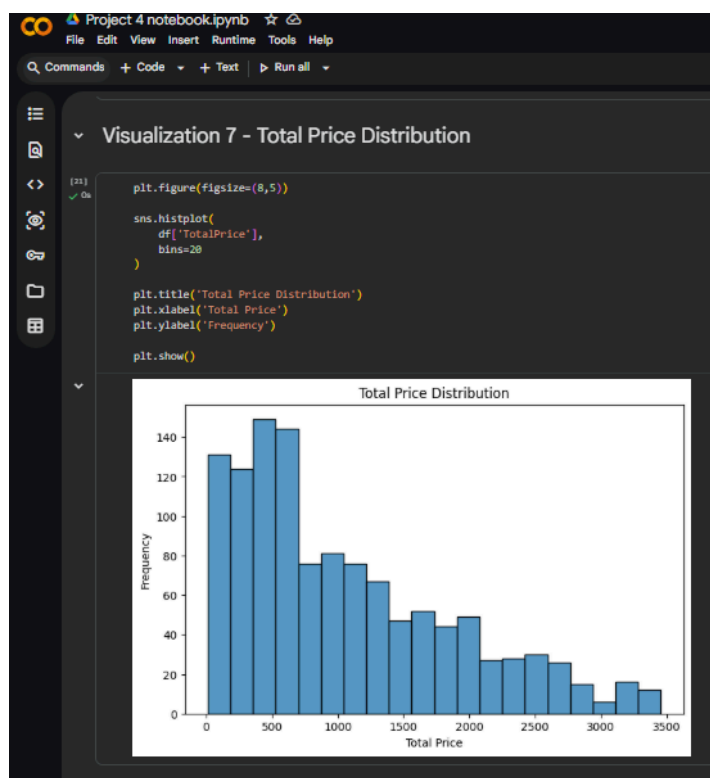
Revenue by Product



Revenue generated by each product was visualized using a bar chart to compare product performance.

The chart highlighted products that contributed the highest revenue to the business.

Total Price Distribution



A histogram was created to examine the distribution of transaction values within the dataset.

The distribution provided insights into common transaction amounts and overall purchasing behavior.

6 Conclusion

This project demonstrated the practical importance of data visualization in data analytics. By using Python, Matplotlib, and Seaborn, raw sales data was transformed into meaningful visual insights.

The project improved understanding of visualization techniques, exploratory analysis, and business intelligence concepts. It also highlighted how graphical representations can simplify complex datasets and support effective decision-making.